

1. when given data relating to affect of air quality:
 - can identify absence of replication as a reason for questioning a scientific claim;
 - **can explain why scientists regard it as important that a scientific claim can be replicated by other scientists;**
 - can identify the outcome and the factors that may affect it;
 - can suggest how an outcome might be affected when a factor is changed;
 - can give an example from everyday life of a correlation between a factor and an outcome;
 - uses the ideas of correlation and cause appropriately when discussing historical events or topical issues in science;
 - **can explain why a correlation between a factor and an outcome does not necessarily mean that one causes the other, and can give an example to illustrate this;**
 - can suggest factors that might increase the chance of an outcome, but not invariably lead to it;
 - can explain that individual cases do not provide convincing evidence for or against a correlation.
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